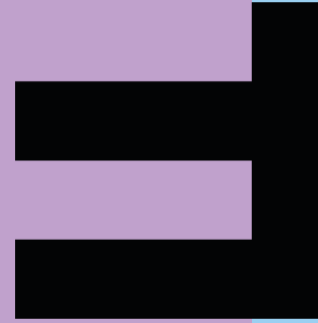


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of Applied Sciences



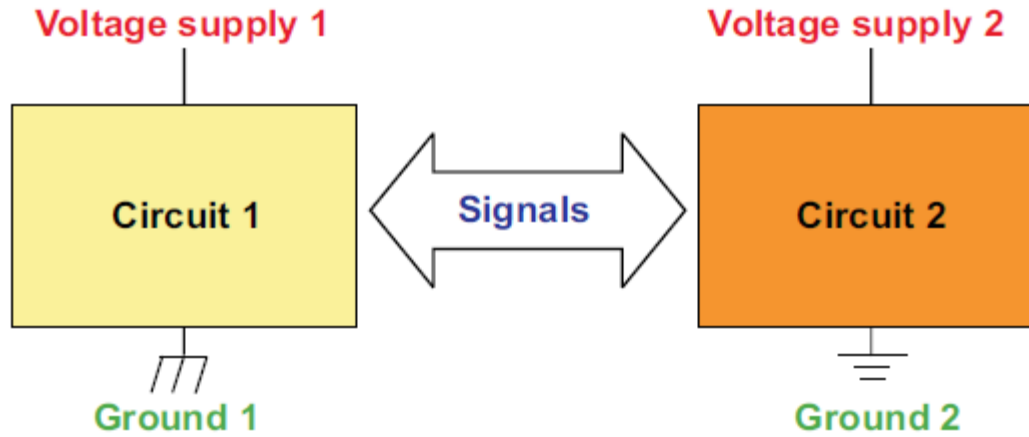
Analog Stuff

13.05.2026

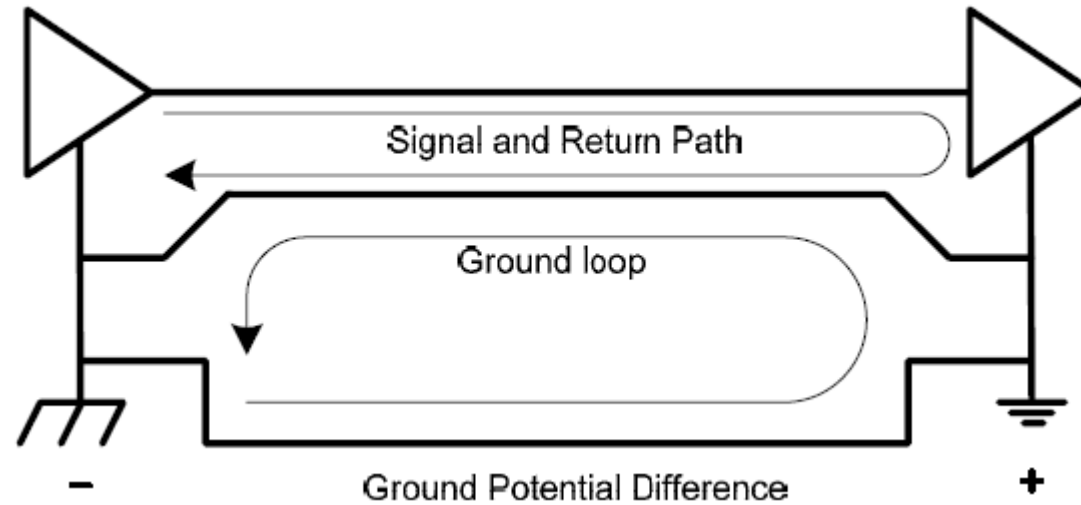
EIT & MEC Bachelor

Horatiu O. Pilsan

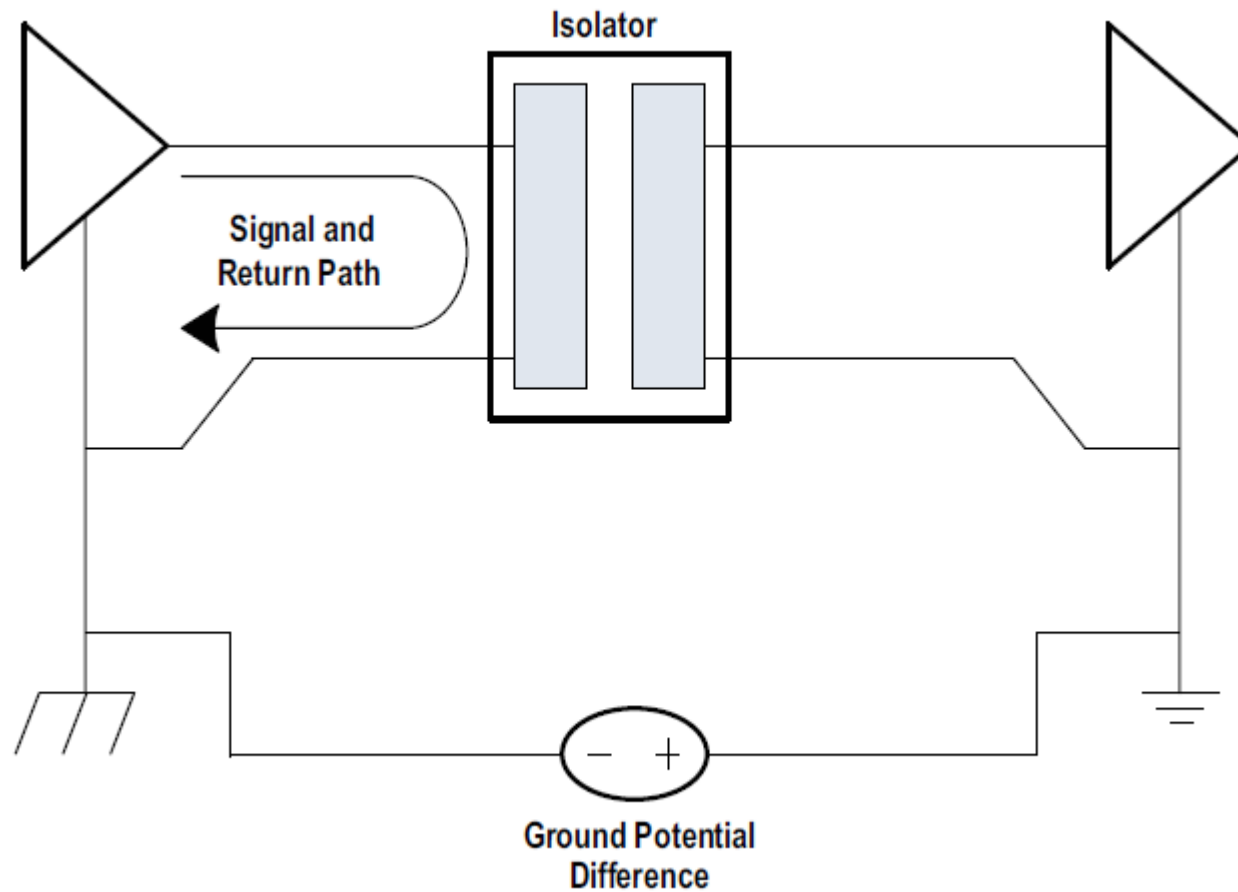
Isolation – Problem



[1], p. 68



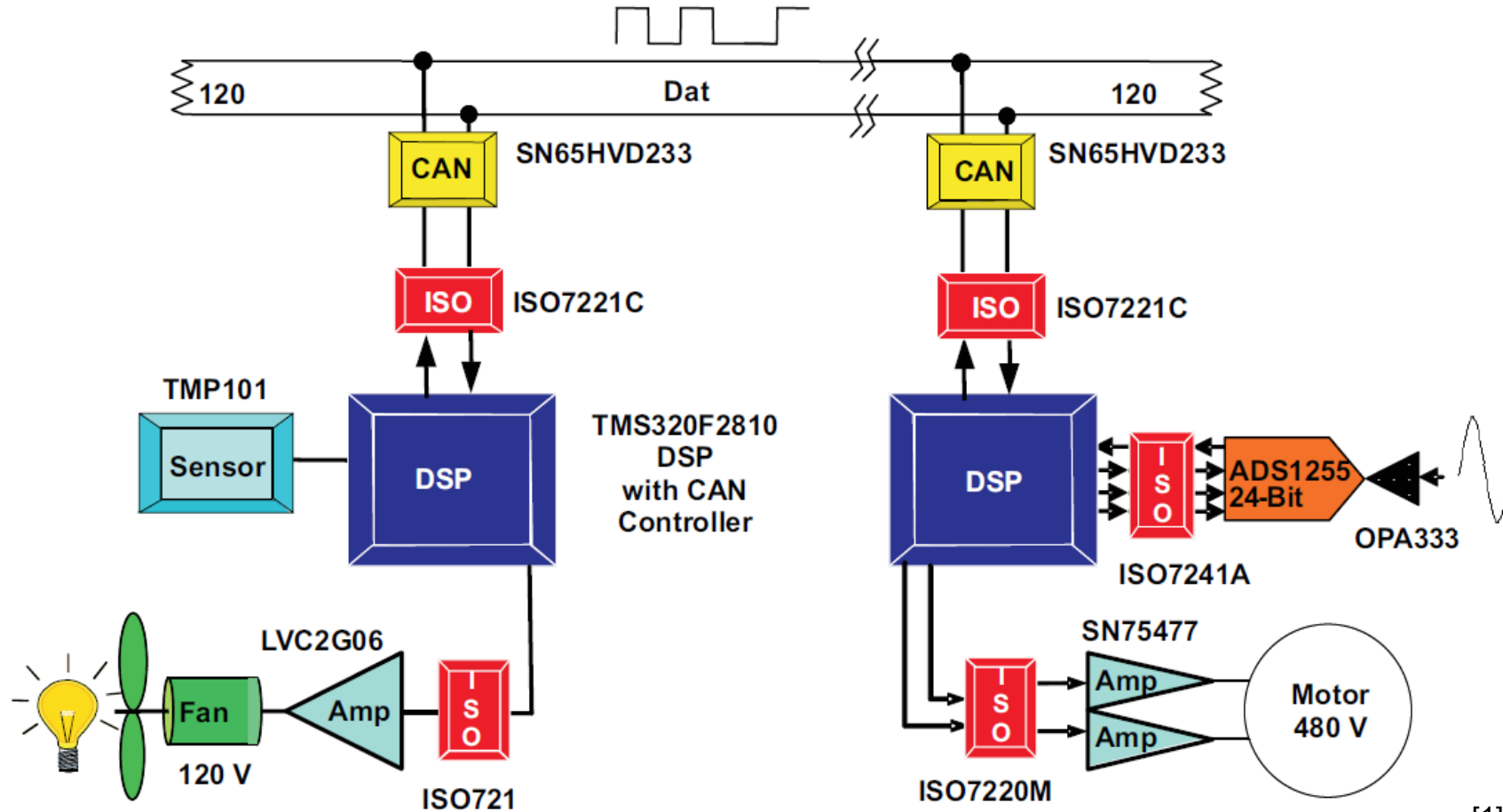
Isolation – Solution



[1], p. 68



Isolation – Application

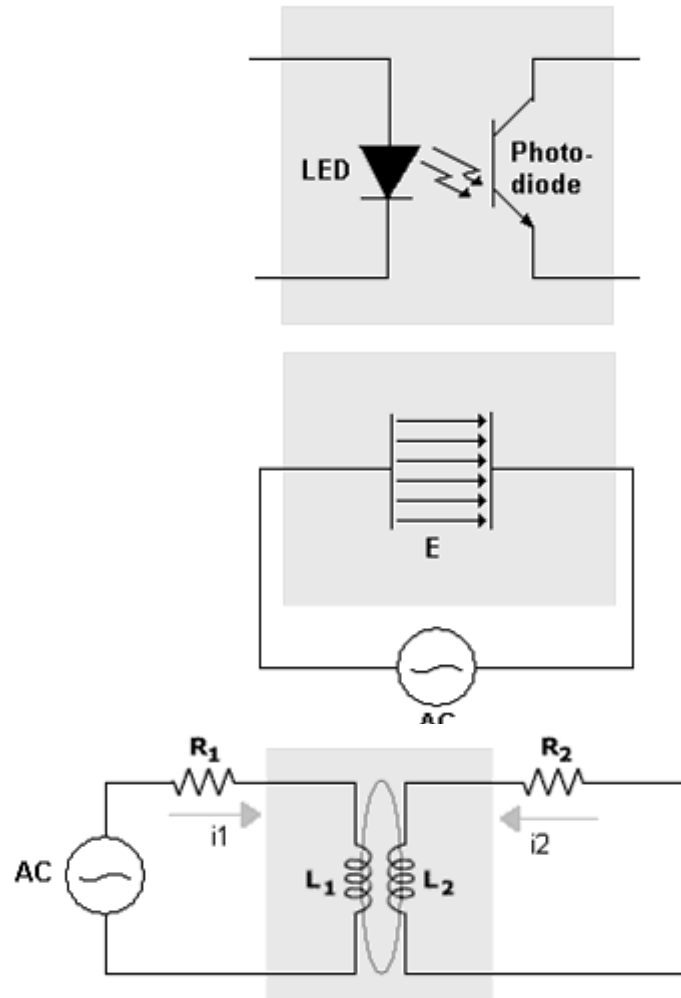


[1], p. 69



Isolation – Types

- Optical
- Capacitive
- Inductive



Isolation – Comparison

	Pro	Con
Optical	Low cost High isolation Noise immunity	Low speed High power
Capacitive	High speed Low power	E noise
Inductive	High speed Lowest power	M noise

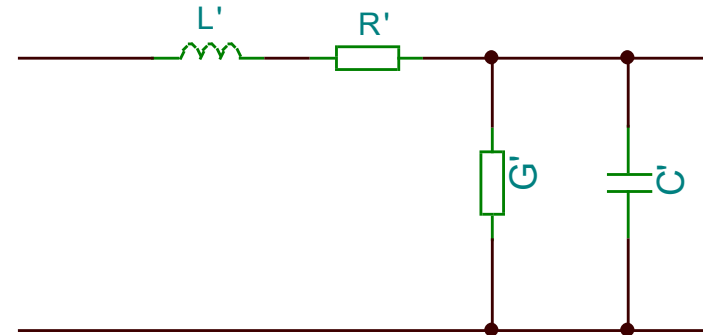


Long Lines Basics I

- Assume the following circuit:



- When is the length relevant?



- Equivalent circuit of an infinitely small part of it:

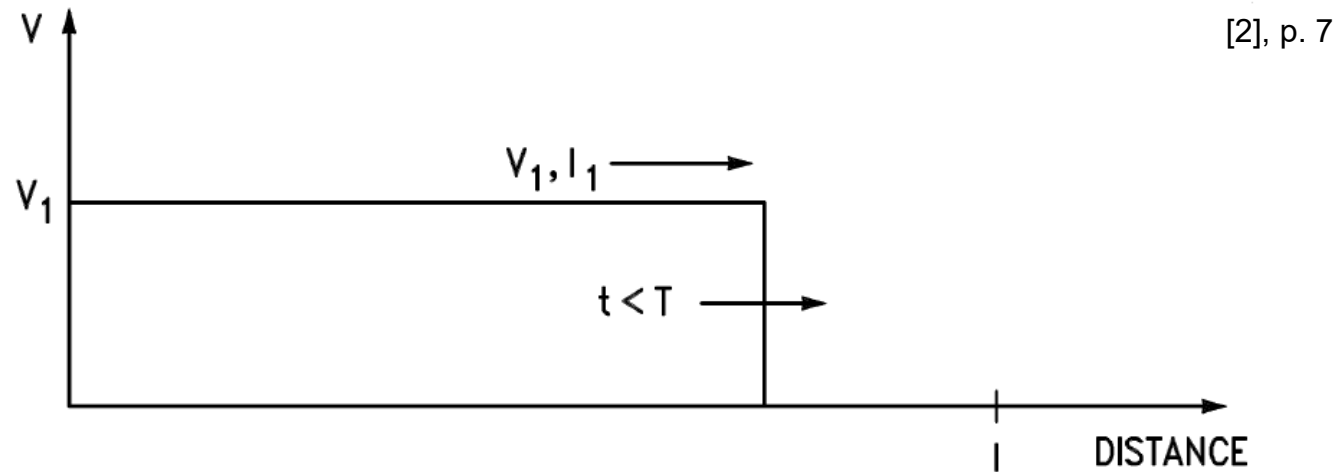


Long Lines Basics II

- Characteristic impedance:

$$Z_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}$$

- Wave propagating:

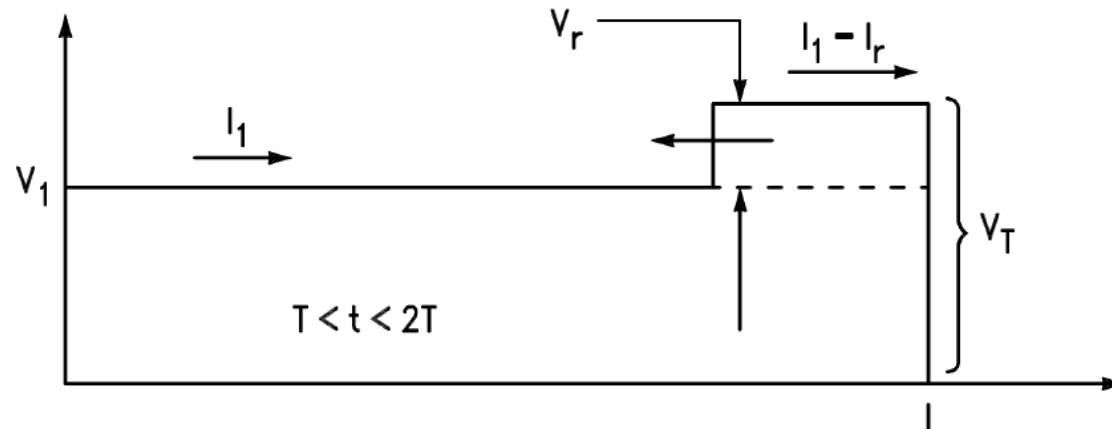


Long Lines Reflection I

- Coefficient of reflection:

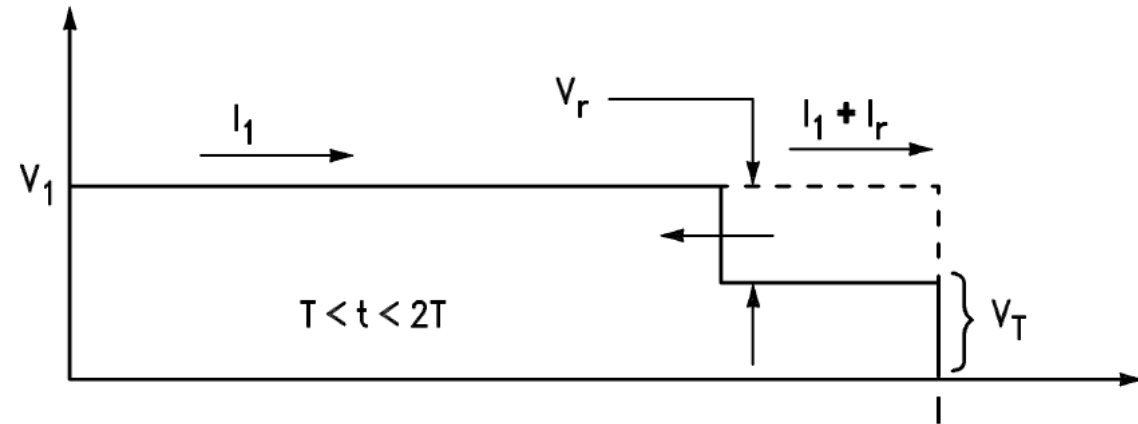
$$\rho = \frac{Z_2 - Z_0}{Z_2 + Z_0}$$

- Reflection for $Z_2 > Z_0$



Long Lines Reflection II

- Reflection for $Z_2 < Z_0$:



- Some special cases:

$Z_2 = Z_0$: $\rho = 0$, wave is not changed

$Z_2 = \infty$: $\rho = 1$

$Z_2 = 0$: $\rho = -1$

[2], p. 7

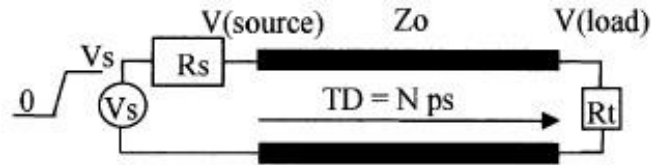
- The initial voltage is given by:

$$V_i = V_s \frac{Z_o}{Z_o + Z_s}$$



The Lattice Diagram I

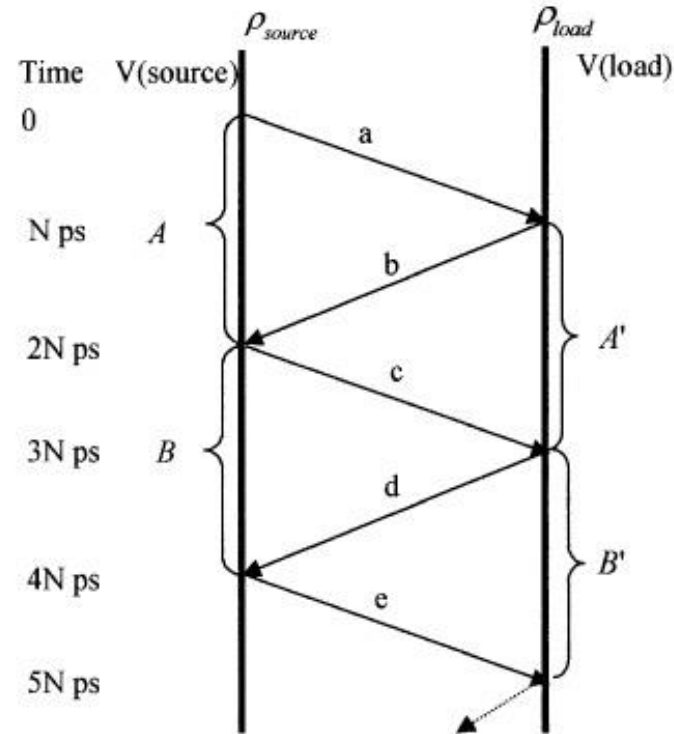
- Lattice diagram:



$$R_s = Z_1$$

$$R_t = Z_2$$

- assume:
- $R_s = 25 \Omega$
- $Z_0 = 50 \Omega$
- $R_t = 300 \Omega$
- $V_s = 3.6 \text{ V}$



$$a = V_{initial}$$

$$b = a \rho_{load}$$

$$c = b \rho_{source}$$

$$d = c \rho_{load}$$

$$e = d \rho_{source}$$

$$f = e \rho_{load}$$

$$A = a$$

$$B = a + b + c$$

$$C = a + b + c + d + e$$

$$A' = a + b$$

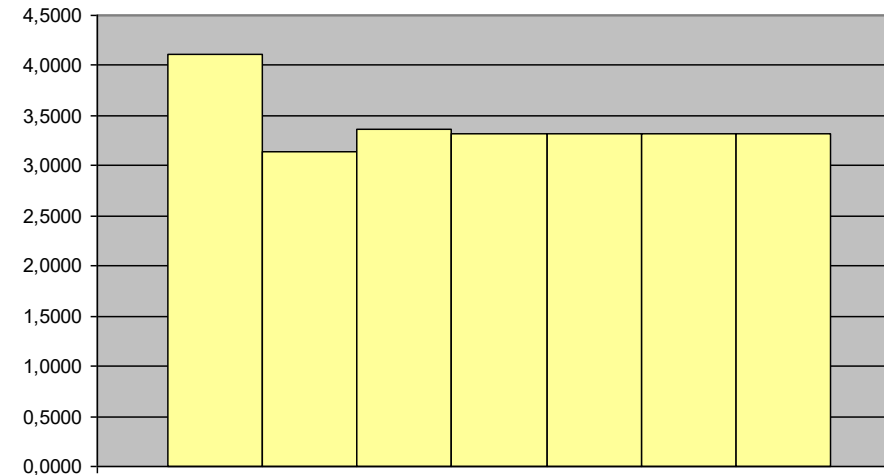
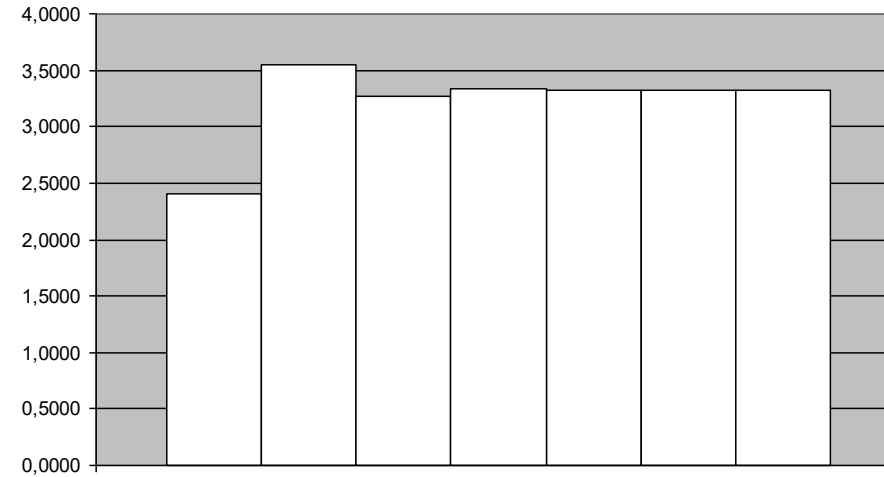
$$B' = a + b + c + d$$

$$C' = a + b + c + d + e + f$$

[3], p. 137

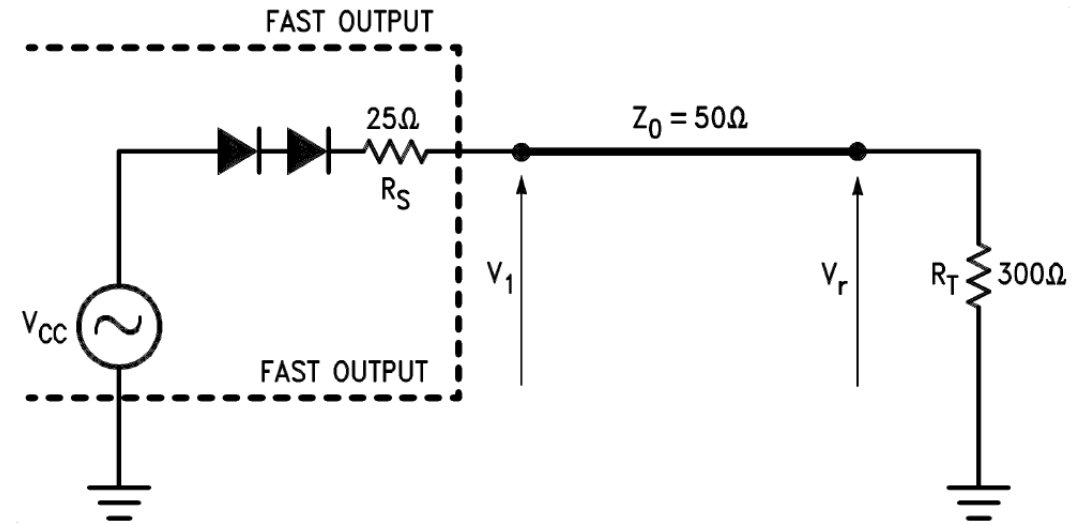
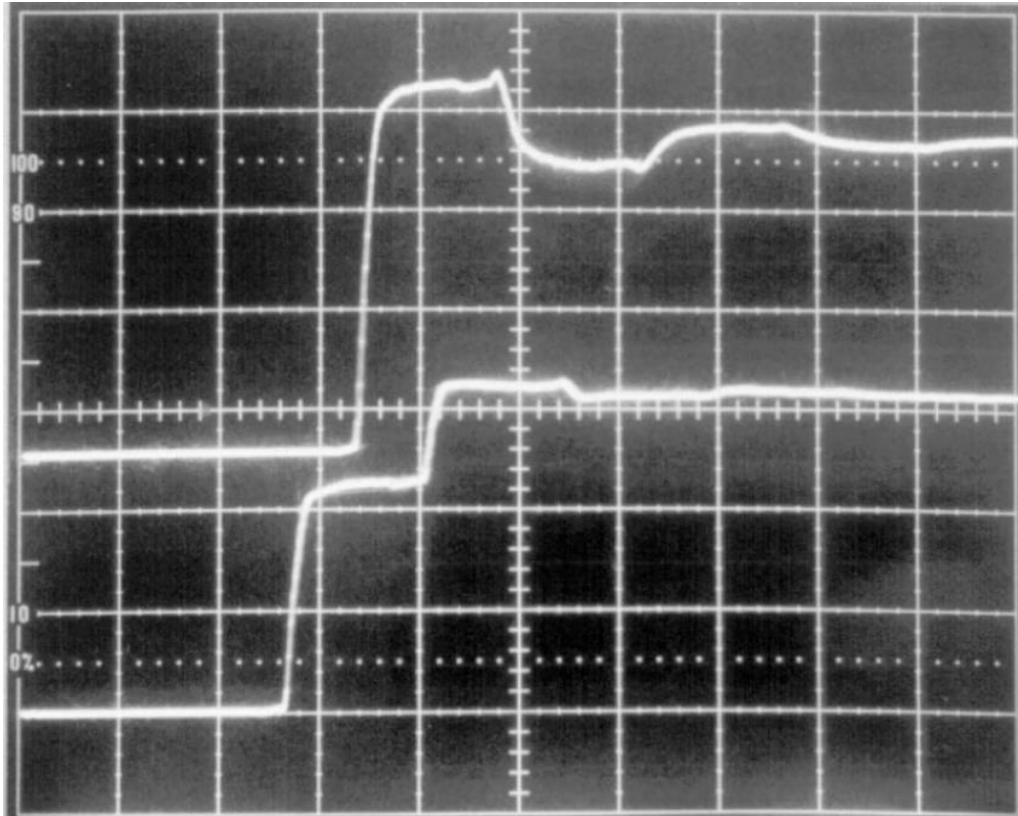
The Lattice Diagram II

R_s	25	a	2,4000	A	2,4000
Z_0	50	b	1,7143	B	3,5429
R_t	300	c	-0,5714	C	3,2707
V_s	3,6	d	-0,4082	D	3,3355
		e	0,1361	E	3,3201
ρ_L	0,7143	f	0,0972	F	3,3238
ρ_S	-0,3333	g	-0,0324	G	3,3229
V_i	2,4000	h	-0,0231		
V_f	3,3231	i	0,0077	A'	4,1143
		j	0,0055	B'	3,1347
		k	-0,0018	C'	3,3679
		l	-0,0013	D'	3,3124
		m	0,0004	E'	3,3256
		n	0,0003	F'	3,3225
				G'	3,3232



Practical Long Lines

- Circuit:
- Practical waveform:



[2], p. 8



Length of Long Lines

- Propagation velocity on a PCB / copper wire, without capacitive input loads:

$$v = 0.4 \dots 0.7 \times c!$$

- Long lines problems have to be considered, when:

$$t_r < 2 \dots 3 \times t_d!$$

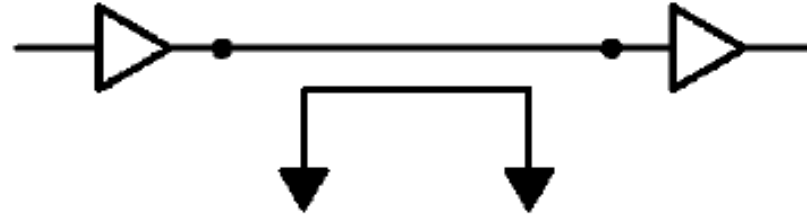
- Assume $t_r \approx 1 \text{ ns}$, let's calculate the relevant length!



Solutions for Long Lines I

- No termination:

Internal clamp diodes effective

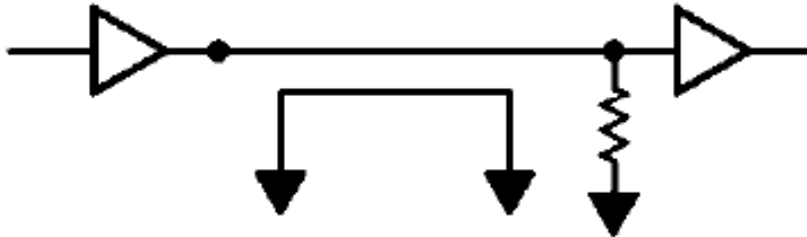


- Parallel termination:

$$R_T = Z_0$$

asymmetric load

high static power consumption

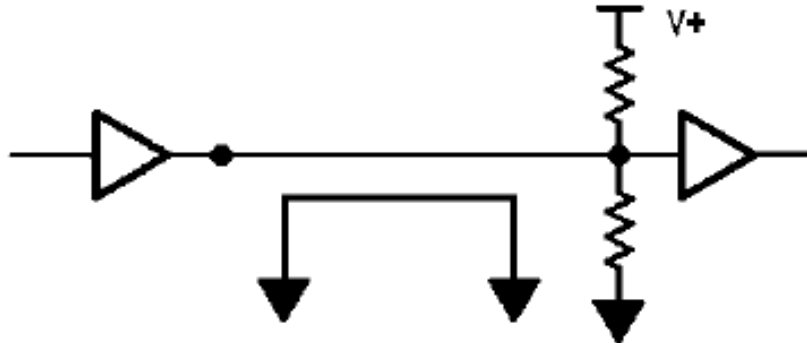


- Thevenin termination:

$$R_T = 2 \times Z_0$$

symmetric load

high static power consumption



Solutions for Long Lines II

- AC parallel termination:

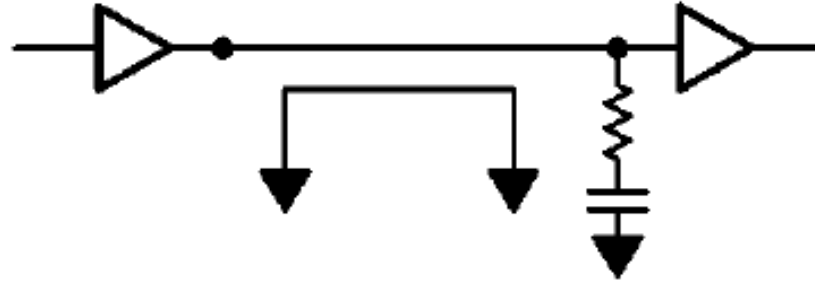
$$R_T = Z_0$$

$$C_T \geq 3 \times t_r / Z_0$$

no static power consumption

practical values:

$$R_T \approx 50 \, \Omega, C_T = 50 \dots 300 \, \text{pF}$$

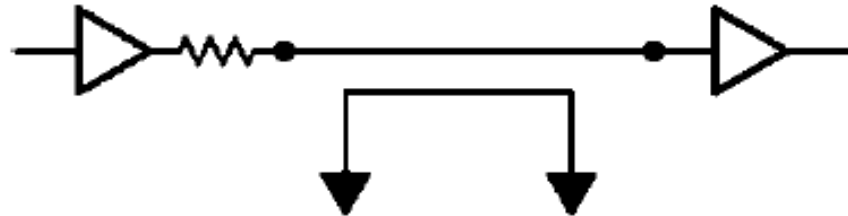


- Series termination:

$$R = Z_0 - R_S$$

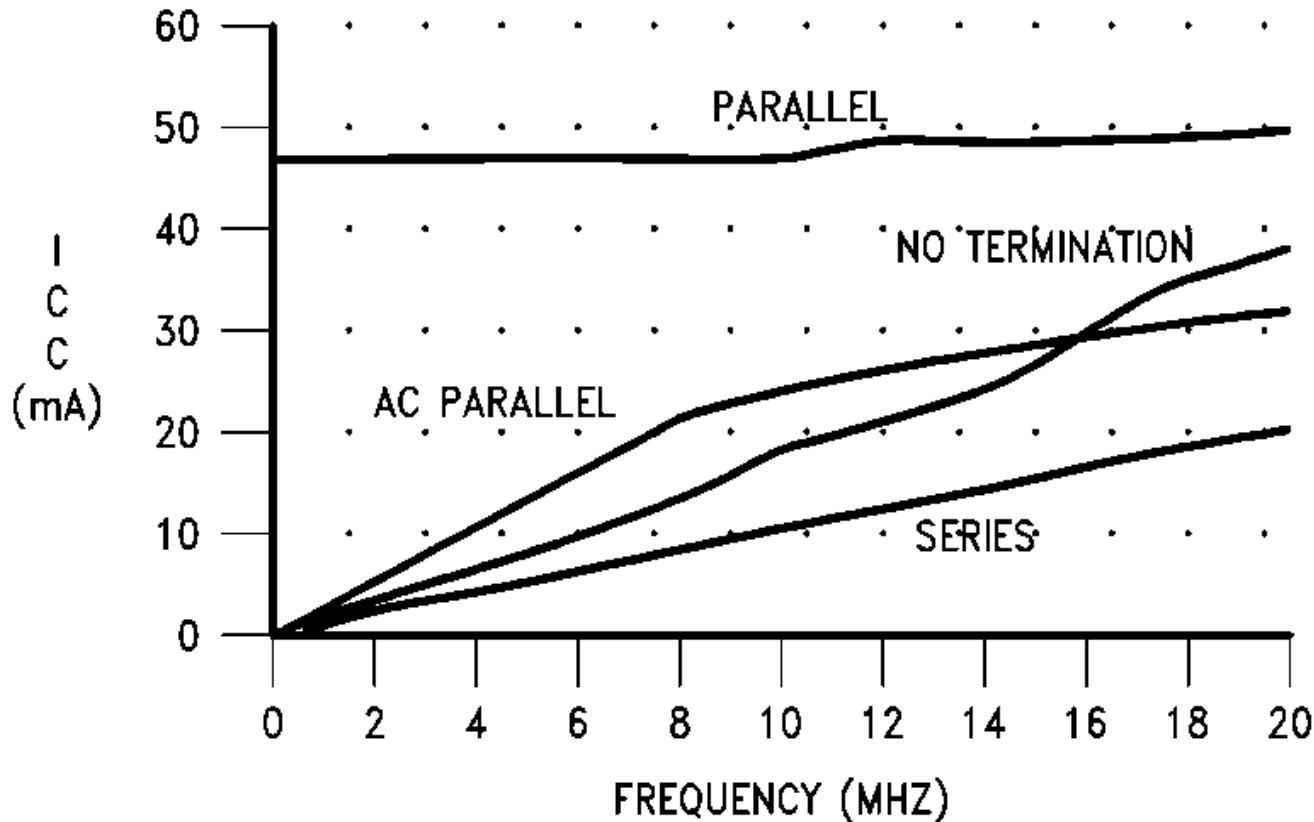
Practical value $R \approx 25 \, \Omega$

Often included in the driver circuit



Line Termination Considerations

- Z_0 can vary due to the capacitive load of the inputs (the number of inputs counts)
- But the order of magnitude is important, not the exact value
- Power consumption is important, see below



[4], p. 4



References

- [1] „Comparing Bus Solutions, Application Report”, Texas Instruments, 2017, document SLLA067B, <http://www.ti.com/lit/an/slla067b/slla067b.pdf>
- [2] „FAST/FASTr Design Considerations”, AP Note 661, Fairchild Semiconductor Corporation, March 1990, Dokument AN009607, <https://docs.rs-online.com/8be7/0900766b80024d5c.pdf>
- [3] Hall, Stephen; Hall, Garrett; McCall, James (2000): High Speed Digital System Design: A Handbook of Interconnect Theory and Design Practices. Hoboken: John Wiley & Sons
- [4] „Terminations for Advanced CMOS Logic”, AP Note 610, Fairchild Semiconductor Corporation, Feb. 1998, Dokument AN010218. <https://docs.rs-online.com/7c92/0900766b80025178.pdf>



Analog Stuff

13.05.2026

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